

RADHEY KALRA

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Full-stack developer and open source contributor focused on Linux systems programming, self-hosted infrastructure, and Audio/ML pipelines.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Model Institute of Engineering and Technology

Aug. 2023 – May 2027

Jammu and Kashmir, India

Schooling (Class I – XII)

Air Force School, Jammu

2011 – 2023

Jammu and Kashmir, India

OPEN SOURCE EXPERIENCE

Linux Kernel (hp-wmi) · Platform-driver-x86 subsystem

[08ecf6d](#) | [0687673](#) | [dcabd63](#) | [b461a88](#)

- Added HP Victus 15-fb0xxx (board 8A3D) fan-control support to the upstream Linux hp-wmi driver by reusing its existing Victus WMI control path.
- Refactored hp-wmi to select thermal-profile and fan-control features from board-specific data instead of hardcoded model checks.
- Tested dual-channel PWM fan control on an HP Victus 16-s0xxx and provided the Tested-by sign-off for the upstream commit.

Tokscale · AI coding token analytics

[bcd4c22](#) (PR #998) | [ab9f226](#) (PR #1000)

- Added first-class standalone Cline CLI and Kimchi Coding session ingestion across parsing, scanning, CLI/TUI filters, frontend metadata, and documentation, preserving token, cost, model, workspace, and cache data.
- Corrected Grok usage accounting and model attribution across parent/child agent sessions with process-generation scoping, deduplication, cache-safe behavior, and dual-source discovery; merged after passing 1,300+ core tests.
- Contributed to a developer analytics project with 4.8K+ GitHub stars and 390+ forks.

ly · Lightweight Linux/BSD display manager

[GitHub commit](#) | [Codeberg commit](#)

- Proposed and implemented an optional big-clock digit outline for ly, improving readability over animated backgrounds while preserving the default appearance; tested with a ReleaseSafe Zig build and a live greeter configuration.
- Contributed to a project with 7.4K+ GitHub stars and 350 forks on its GitHub mirror.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Go, C/C++, Rust, SQL, Bash, PowerShell, Vala

Frameworks & Platforms: SvelteKit, React, Astro, Tailwind CSS, React Native, Bun

Developer Tools: Git, Docker, CI/CD, Cloudflare, Vercel, Kubernetes, Terraform

Databases & Cloud: Convex, PostgreSQL, MySQL, AWS, GCP

Domains: Full-Stack Development, Linux Servers, Self-Hosting, Networking, Audio Processing

PROJECTS

BoxBox

Nov. 2025 – Present

Self-hosted file management | [Website](#) | [GitHub](#) | [Docker](#)

- Made self-hosted file management practical on lightweight hardware with multi-mount browsing, chunked/resumable uploads, live job progress, media preview/streaming, and in-browser code editing, all in a 31MB single binary.
- Eliminated deployment friction with an embedded UI and a one-command container image for homelab and NAS setups.
- Reached 300+ GitHub stars and 13K+ downloads, confirming demand for a lighter alternative.

U-Net Audio Filter

Jul. 2025 – Present

Audio enhancement / background noise removal | [GitHub](#)

- Improving audio denoising quality beyond existing open-source baselines with a custom U-Net architecture and an audio preprocessing pipeline.

HPomen Linux (Victus Control)

Mar. 2026 – Jul. 2026

Reverse-engineering HP OMEN control utility behavior on Linux | [GitHub](#) | [AUR](#)

- Built a Linux-first OMEN/Victus control suite in **Vala** with a GTK4 monitor app, tray companion, D-Bus helper, and probe CLI.
- Reverse-engineered **HP WMI/sysfs** behavior to validate profile/fan controls (**auto/max**) and extend **hp-wmi** with capability-gated safety fallbacks.
- Submitted a Linux kernel **hp-wmi** patch accepted into **platform-driver-x86**, adding Victus 15-fb0xxx board support for fan-control feature data; also provided **Tested-by** validation for another **hp-wmi** fan-control patch on real hardware.

Fast Blob Storage (FBS)

Apr. 2026 – Present

S3-compatible object storage | [Website](#) | [GitHub](#) | [Docker](#)

- Led a team of 5 contributors to design and build a single-node object store, coordinating API, storage, dashboard, and deployment work across a single-binary and containerized release.
- Matched MinIO's core API (SigV4 + bearer auth, multipart uploads, management API) at 12–14× less memory using zero-copy **sendfile()** egress instead of distributed-cluster overhead.
- Grew to 50+ GitHub stars, proving lightweight object storage can compete without sacrificing performance or compatibility.

Additional Projects

- **QckPages** — End-to-end HTML page publishing platform ([Live](#) | [GitHub](#))
- **rsclip** — Rust/Wayland clipboard manager with OCR and secrets management ([GitHub](#) | [AUR](#))
- **Dots-Niri** — One-command NiriWM rice for Arch-based installs ([Website](#) | [GitHub](#))
- **HowTo** — Tutorials for practical hardware/software walkthroughs built with Astro ([Live](#) | [GitHub](#))
- **Pico32** — C++/PlatformIO serial TUI for ESP32 Wi-Fi/BT pentesting ([Website](#) | [GitHub](#))
- **TL;DR** — Electron app summarizing WhatsApp Group chats via OpenRouter models ([GitHub](#))
- **Hotel Booking** — Svelte/Express/MySQL hotel booking platform with auth and admin flow ([GitHub](#))
- **Tenant Manager** — React Native app for tenant management and electricity tracking ([GitHub](#))

POSITIONS OF RESPONSIBILITY

Cloud & DevOps Lead

Aug. 2025 – Present

Google Developer Groups on Campus & AWS Cloud Club

MIET, Jammu

- Lead cloud and DevOps initiatives across Google Developer Groups on Campus and AWS Cloud Club through workshops, technical sessions, and collaborative projects on GCP, AWS, containerization, and infrastructure automation.
- Ran Google Cloud sessions for 100+ students and built a tutorial platform with 20+ lab walkthroughs, notes, and videos for Google Cloud Study Jam ([Live/GitHub](#)).

ACHIEVEMENTS

IDE Bootcamp Edition 3 – Team Leader, 2nd Place

Apr. 2026

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

GNDEC, Ludhiana, Punjab

- Led a team to 2nd Place building an AI-powered SaaS product for automated presentation generation from user-uploaded templates and documents with lower cost and no hallucinations.

IDE Bootcamp Edition 2 – Team Leader

Feb. 2025

Innovation, Design, and Entrepreneurship (IDE) Bootcamp

IIT Jammu

- Led a team in using design thinking and Lean Canvas to develop a sports matchmaking and social networking app.

Hackathon Team Leader

Internal Hackathons & Smart India Hackathon

MIET, Jammu

- Led multidisciplinary teams to the top 9 in college both years through collaborative problem-solving and rapid prototyping.

CERTIFICATIONS & PROGRAMS

Google Cybersecurity - Coursera | [Certificate](#)

Cisco Cybersecurity Operations Fundamentals - Coursera | [Certificate](#)

Palo Alto Networks Cybersecurity - Coursera | [Certificate](#)

EC-Council Cybersecurity Attack and Defense Fundamentals - Coursera | [Certificate](#)

NPTEL Ethical Hacking - IIT Kharagpur | [Certificate](#)

NPTEL Deep Learning - IIT Ropar | [Certificate](#)

IBM SkillsBuild: AI Agent Architect | [Certificate](#)

IDE Bootcamp (AICTE / MoE) | [Certificate](#)